CURRENT AND FUTURE GENERATION

Electricity Transmission Systems

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- National and regional grid systems connecting generators with wholesale customers are generally just as important as electrical power generation.
- Investment in these is often on a similar scale to generation capacity.
- New technology is enabling transmission at high voltages over long distances without great losses.
- Transmission system operators (TSOs) have responsibility for the quality of power supply.
- Where national energy policies prioritise security of supply, the role of TSOs is to achieve operational reliability, from diverse sources with different characteristics.

Countries with well-developed electricity infrastructure have established grids run by transmission system operators (TSO) to convey power to distribution systems where it is needed. Where generating plants can be located close to load centres, these are less important than where the plants are remote, as with many hydroelectric plants and wind farms. Lower voltage can be used. At higher voltages, eg 500kV and above, transmission losses over hundreds of kilometres are much reduced. At ultra-high voltages (UHV), e.g. 1000 kV AC or 800 kV DC, losses are further reduced (e.g. to 5% over 1000 km or 3.5% for HVDC) but capital requirements are greater. New plans are for 1100 kV DC and 1050

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kV AC links. In Germany consideration is being given to converting some existing AC lines to DC to increase their capacity.

Transmission losses are often around 6%, though a world average is given as 8%. In the USA the estimate is about 6%, or 250 TWh per year, worth some $20 billion. The EU loses 6% but the UK figure is 8%. China is working to reduce transmission losses from 6.6% in 2010 to 5.7% in 2020, Japan in 2013 had 5% losses and South Korea 3%. In India transmission losses in 2011 were 222 TWh (21%) and in 2013 were 18%, mostly attributed to theft. Some countries are higher. (International Energy Agency statistics)

Wholesale distribution companies ('discos') step the voltage down with transformers, eventually to domestic voltage, and market the electricity.

Transmission grids are normally alternating current (AC), which can readily be transformed to higher or lower voltages. Increasingly, direct current (DC) links are being used for particular projects, particularly undersea cables linking countries or linking offshore wind farms to onshore grids via converter stations. Also high-voltage DC links (HVDC) are becoming more important for long-distance efficient transmission.

Typically 132 kV or higher voltage will connect power plants and provide the backbone of a grid system, while 66 kV, 33 kV or 11 kV may connect renewable generation sources such as wind to it. Distribution is at 400 volts and eventually less.

In a synchronous grid such as western Europe, all generators are in phase, allowing AC power to be transmitted throughout the area, connecting a large number of electricity generators and consumers and potentially enabling more efficient electricity markets and redundant generation capacity. In May 2014 the power grids and exchanges in southern and north-western Europe were connected, covering about 70% of European customers and with annual consumption of almost 2400 TWh. The common day-ahead power market created through the physical and financial integration of the two regions extends from Portugal to Finland. This is expected to lead to a more efficient utilization of the power system and cross-border infrastructures, as a result of a better harmonization between energy markets. It is expected that electricity markets in the Czech Republic, Slovakia, Hungary and Romania will join similarly and then link to the rest of Europe. Poland is partially integrated with north-western region in Europe through a subsea line to Sweden. Italy's possible integration will depend on

Switzerland's discussions with the European Union on connecting power systems.

Sometimes AC grids are connected by high voltage DC (HVDC) links, using voltage-source converters (VSCs). HVDC allows connection of asynchronous AC systems. Over 300 GW of new HVDC transmission capacity is expected to be added to world grids by 2020, two-thirds of this in China to connect inland renewable (especially hydro) sources to coastal load centres. In July 2016 Siemens received its first order for 1100 kV converter transformers for the 3200 km 12 GW Changji to Guquan HVDC link in China, expected in operation at the end of 2018.

One major issue for many countries setting out to add nuclear capacity to their infrastructure is the size of their grid system. Many nuclear power plants are larger than the fossil fuel plants they supplement or replace, and it does not make sense to have any generating unit more than about one tenth the capacity of the grid (maybe 15% if there is high reserve capacity). This is so that the plant can be taken offline for refueling or maintenance, or due to unforeseen events. The grid capacity and quality may also be considered regionally, as with Jordan for instance. In many situations, as much investment in the grid may be needed as in the power plant(s).

In Europe, the power transmission system operating body ENTSO-E, comprising 41 TSOs from 34 countries, has assessed the ability of Europe's grid networks to become a single internal energy market. This will require some $128 billion in new and upgraded power lines on order to meet the EU's renewables and energy market integration goals. In its 2012 Ten-Year Network Development Plan it identified 100 power bottlenecks standing in the way, with 80% of them relating to the challenge of integrating renewable energy sources such as wind and solar power into national grids. Much of the European investment needs to be on refurbishment or construction of about 51,000 km of high voltage power lines and cables, to be clustered into 100 major investment projects dealing with the main bottlenecks. One goal (set in 2002) is to have a level of interconnection for each country at least equivalent to 10% of its generating capacity, to achieve trans-EU electricity infrastructure. This was far from being achieved in 2013, but the above investment will bring it about for all EU countries except Spain. One bottleneck is being addressed by building a 1400 MW HVDC link 65 km across the Pyrenees to double the Spain-France capacity, the longest underground HVDC link in the world, costing €700 million. A planned second link undersea will increase interconnection to 5000 MW about 2020.

An ENTSO-E 2013 research project involved 20 partners from 12 countries to redefine reliability in an increasingly interconnected, renewables-dominated system. The GARPUR (Generally Accepted Reliability Principle with Uncertainty Modelling and Through Probabilistic Risk Assessment) project focuses on the optimal balance between the costs of providing a secure supply of electricity and the socio-economic costs of power interruptions in an increasingly complex system. This approach takes into account the probabilities of failures based on weather dependence, maintenance history, and real-time conditions. It considers uncertainties in both generation and load forecasts, and the flexibility provided by the demand side, energy storage and distributed renewables. It enables governments, regulators and TSOs to put a price on security of supply and minimise the cost of achieving it.

Transmission capacity from producers needs to be sufficient for peak output from them. Hence per MWh delivered to wholesalers this is three or four times more costly for intermittent renewables than for base-load plants. In Austria the grid availability and line loss charges for 2015 are set so as to be about €3.50/MWh for renewables.

Germany is a prime example of the need for increased transmission capacity, having its traditional fossil fuel and nuclear power generation plants in the south, with lines spreading from there to the rest of the country, while wind power sources are along its northern Baltic coast. Hence its existing north-to-south to lines have become bottlenecks, incapable of transmitting sufficient wind-generated power from the north to replace closed capacity in the south.

The TSOs said their analysis showed that extending the grid by only 1.3 percent enables the addition of 3 percent generation capacity and the integration of 125 gigawatts of renewable energy sources -- all at a cost of 2 cents per kilowatt-hour for electricity consumers over a 10 year span. "Cumbersome permit-granting procedures and a lack of public acceptance for power lines are presently the most relevant obstacles" facing the efforts. Hence ENTSO-E proposes that each EU member state should designate a single competent authority responsible for the completion of the entire permit-granting process, which would not exceed 3 years.

Another goal of the EU's grid infrastructure efforts is reducing the "energy island" status of Italy, Iberian Peninsula, Ireland, UK and Baltic states. This will be addressed by the upgrades, while reducing the total generation costs by about 5%.

The planned 1.4 GWe Nordlink HVDC connection between Germany and Norway has great potential to link northern Germany's solar and wind capacity with Norway's hydro from 2020, providing crucial back-up for Germany and enabling surplus wind and solar power to be exported north. The 620 km link west of Denmark is expected to cost $2.8 billion total. However, Norway is insisting that the deal take account of the fact that its hydro capacity is dispatchable, and that it be part of any capacity market which rewards this attribute in backing up Germany's intermittency. Germany is reported to see the link as vital to its plans for phasing out dispatchable nuclear power in 2022. Norway's Stattnett will own 50%, Germany's TenneT TSO and a state-owned bank KfW Group will own 25% each. Norway produces about 95% of it electricity from hydro. It already has transmission links with Sweden, Denmark (1700 MWe, with another 700 MWe HVDC planned), and Netherlands (NorNed, 700 MWe), and is building a 730 km, €2 billion HVDC link to UK (1.4 GWe NSN link, due on stream in 2021). The NSN project has been chosen as one of the European Commission's projects to help establish an integrated EU energy market.

An EC-sponsored Booz study in 2013 supported ENTSO-E's plan to increase transmission by 40% by 2020, but said this rate must be maintained to 2030. "Around 90% of the benefits are achievable even if only half the desirable increment in transmission capacity is achieved, even without demand side reduction," it said. The study said that integrating EU electricity markets more closely could yield up to €40 billion per year in benefits by 2030, and coordinating renewables investment could add €30 billion per year to that. Improving demand-side responses with smart grids could add up to €5 billion/year, and sharing balancing costs could add up to €0.5 billion/ year, the study says, bringing total potential benefits to €75.5 billion/year by 2030.

In a world perspective, France's RTE estimates that $700 billion investment in the 16 largest grids handling 70% of world electricity is required over the ten years to 2022, partly on account of integrating renewable sources. The 16 grids have 2.2 million km of lines. RTE itself plans to invest $19 billion by 2020. In developed countries grid development is slow due to the approval process and public opposition.

Dispatchability

The main challenges for grid management are frequency and voltage control, in the course of supplying demand which is constantly varying. This means that there must be dispatchable capacity available for TSOs to call upon. Traditionally this is dispatched in merit order, ie according to lowest marginal cost. However, with the establishment of preferential access for intermittent renewables, coupled with relatively high feed-in tariffs or other arrangements, this is increasingly compromised. Where there is large intermittent renewable capacity connected to the grid, supply from it may meet much of the demand or even exceed it at times, which means that reliable, low-marginal cost capacity is then shut down. Since that kind of plant is often high-capital, low-running cost equipment, its economic viability is subverted.

Grid management authorities faced with the need to be able to dispatch power at short notice treat wind-generated power not as an available source of supply which can be called upon when needed but as an unpredictable drop in demand. In any case wind needs about 90% back-up, whereas the level of back-up for other forms of power generation which can be called upon on demand is around 25%, simply allowing for maintenance downtime. Some discussion on the integration costs of renewables is in the companion WNA paper on Renewable Energy and Electricity.

Where there is substantial renewables input on occasions, there are increasing calls for capacity payments, or capacity remuneration mechanisms (CRM) – provision to pay utilities to keep dispatchable capacity available and, in the medium term, to encourage investment in such. Germany is the country which has seen most gas-fired plant made uneconomic by its Energiewende provisions to encourage renewables, and it is proposing two types of capacity payments: one customer-based as in France, and one with a central buyer, as planned for UK. At the start of 2014, half of EU countries had or were planning some kind of capacity market. In the UK system, capacity requirement will be defined administratively according to TSO forecasts, and the price determined by auction. In the French system, capacity requirement is defined by decentralised demand in the retail market, and price is determined by tradeable certificates. The central system has the effect of socialising investment risks. The first UK auction for 2018-19 capacity will be in November 2014. Eurelectric has called for CRMs to be market-based rather than state aid, and to be technology-neutral and non-discriminatory, and regionally linked.

Maintaining dispatchable power is increasingly difficult with the advent of high renewables capacity. But the costs of failing to meet demand are very high. The value of lost load (VOLL) is estimated at

50 to 350 times the price of a delivered kWh. Hence a capacity margin must be maintained to cater for unexpected surges in demand and the variability of renewables input.

Ancillary services: voltage and frequency control

One of the basic functions of a TSO is to ensure that voltage in distribution networks and frequency do not depart significantly from the criteria set. It also needs to control the power flow (network loading) and cater for unusual disturbances. A TSO will often contract for these ancillary services in advance.

Frequency control ancillary services (FCAS) are fundamental, and there are two types in a grid: regulation control smooths out the routine minor load or generation variation; contingency control is correction of the generation-demand balance to avoid large frequency excursions in the network arising from a major interruption in supply. The first is used continually and centrally controlled, the latter only occasionally and more locally.

In the UK, the National Grid has a statutory duty to maintain the frequency in the range of 49.5–50.5 hertz and usually maintains it at 49.8 Hz to 50.2 Hz. In Australia the automatic generation control maintains frequency at 49.85 to 50.15 Hz. Elsewhere, 0.25 Hz variation can be tolerated. Regulation control is by adjusting power output from generators. Contingency control may require more major changes to generation, or load shedding, depending on timeframe. In France, 49.2 Hz is designated the security level, and below 49 Hz load shedding occurs.

Rapid frequency changes are attenuated due to the inertia of the rotating turbine generators in conventional synchronous power plants, this being called instantaneous reserve. In systems with high renewables contribution the electronics of the feed-in inverters can emulate this to some extent as synthetic inertia. Without this, it is necessary to limit the instantaneous penetration from asynchronous sources such as solar and wind. As well as this there is normally emergency reserve or ‘spinning reserve’ equal to the capacity of the largest generator in the system.

Following a complete blackout in the whole state of South Australia in September 2016 due to loss of voltage and frequency control when most power was coming from wind farms, the Australian Energy Market Operator (AEMO) will require at least two synchronous generators always to be online in the state (as well as

keeping some reserve capacity from interstate). The third AEMO interim report on the incident says: “System strength … primarily depends on the quantity of nearby online synchronous generators.”

A key metric is the rate of change of frequency (RoCoF). Small plants are set up to survive only small RoCoF, e.g. 0.5 Hz/s, and with any more they trip out (disconnect). Large generators have to survive up to 3 Hz/s RoCoF before tripping out. A major blackout in South Australia in September 2016 resulted after the RoCoF reached 7 Hz/s.

In Japan, due to damage caused by the March 2011 Tōhoku earthquake, Tepco’s frequency dropped to 48.44 Hz in just over a minute, but load shedding of 5570 MW followed quickly by another 135 MW in the immediate area avoided a system blackout. Frequency was recovered in about five minutes with increased generation (though the loss of 9100 MWe of supply took a week to rectify after rolling blackouts).

Early in 2016 the UK’s National Grid got a strong response to a tender for 200 MWe ‘enhanced frequency response’. It offered four-year contracts for capacity able to provide 100% active power output in a second or less of registering a frequency deviation. Some 888 MWe of battery capacity was offered, 150 MWe of interconnection, 100 MWe of demand-side response and 50 MWe of flywheel capacity. In September the winning bids for enhanced frequency response were announced – the 64 projects being from 10 MWe to 49 MWe and costing £66 million in total. All but three involve battery storage. The winning bids ranged from £7 to £12 per MWh of enhanced frequency response.

In Europe, there has been a proposal for TSOs to allow more variation in frequency, e.g. from 50 down to 47.5 Hz for extended periods, so that intermittent renewable sources can better be accommodated. Increased contribution from renewables is called for by some EU governments, but in the case of Germany an ancillary services study to 2033 suggests that frequency control can be managed. ENTSO-E says that that the proposal for greater flexibility is to resolve "cross-border network issues and market integration issues," one of which requires "facilitating the targets for penetration of renewable generation.” Currently, variation of up to 1 Hz is allowed briefly. The Western European Nuclear Regulators Association (WENRA) has said this proposal has "the potential to negatively affect nuclear safety" because "the definition for the range of frequency and voltage is too large." In addition, variability accelerates the ageing of some plant components, especially electric motors. ENTSO-E data shows that increased penetration of

renewables is related to a sharp rise in the number and duration of frequency events.

Under technical and design specifications for nuclear safety the lowest frequency allowed for safety-related equipment is 48 Hz and a frequency below that means, for example, that a coolant pump might operate too slowly. Furthermore, nuclear legislation of several WENRA countries does not allow nuclear plants to participate in frequency control or load following, as proposed by ENTSOE-E.

Voltage control ancillary services are related to keeping power flow within the physical limitations of equipment. One method of voltage control is where generators absorb reactive power from the electricity grid or generate reactive power onto it, and control the local voltage accordingly. This can also be done by high-inertia rotating stabilisers in the grid system. In the EU the permanently permissible range of generator voltage variation is defined from 95% to 105% of rated voltage for up to 15 minutes. For a limited time, generators ought to be capable of operating in a voltage range from 92% to 108% of rated voltage in order to compensate for TSO issues, basically to ensure synchronous operation of the grid and support the system when local voltage problems occur (e.g. to avoid voltage collapse). At the transmission system connection point for distribution, the voltage is allowed to vary by 10%. In Germany several new means of providing increased reactive power in the grid are being explored, including phase shift transformers, and some redispatch can be used. Also provision of reactive power via inverter stations of planned HVDC lines is envisaged.

Controlling voltage and frequency coupled with rapid ramp up and down are the main challenges arising from the increasing share of solar and wind renewables in any grid. Sufficient dispatchable synchronous generation capacity must be connected in order to provide inertia to maintain frequency. Asynchronous input from wind and solar PV is not on its own able to provide the required controls to ensure system security, giving rise to the need for other measures.

Some synthetic inertia can be provided from the electronics of the feed-in inverters from wind turbines or more reliably, synchronous condensers can provide enough real inertia to stabilize the system. These are high-inertia rotating machines that can support the grid network in delivering efficient and reliable synchronous inertia and can help stabilize frequency deviations by generating and absorbing reactive power. Some newer wind turbines are directly

coupled and run synchronously at fixed grid-defined rotation speeds, providing some frequency stability, although less total energy output than with DC output.

Synchronous condensers are like synchronous motors with no load and not mechanically connected to anything. They may be supplemented by a flywheel to increase inertia. They are used for frequency and voltage control in weak parts of a grid or where there is a high proportion of variable renewable input requiring grid stability to be enhanced. Adding synchronous condensers can help with reactive power needs, increase short-circuit strength and thus system inertia, and assure better dynamic voltage recovery after severe system faults. They can compensate either a leading or lagging power factor, by absorbing or supplying reactive power (measured in volt-ampere reactive, VAr) to the line. Some generators retired from coal-fired plants are converted to synchronous condensers powered from the grid.

In Germany a highly variable flow from offshore wind farms in the north is transmitted to the main load centres in the south, leading to voltage fluctuations and the need for enhanced reactive power control. The reduced inertia in the entire grid made the need to improve short-circuit strength and frequency stability more critical and was addressed by installing a large GE synchronous condenser at Bergrheinfeld in Bavaria. Following a state-wide blackout, South Australia is installing four Siemens synchronous condensers to compensate for a high proportion of wind input to the grid and reduce the vulnerability to further problems from this.

In the UK, Statkraft plans to install two GE rotating stabilisers to provide stability services to the transmission network in Scotland. These would draw about 1 MWe from the grid and provide synchronous inertia to enable many times that of intermittent renewable input, replacing the role of inertia in fossil-fuel or nuclear plants. The project is among five innovative grid stability contracts awarded by the National Grid electricity system operator in January 2020.

Some definitions of ancillary services include redispatch and curtailment, along with load-following, among other services to achieve reliable operation of the grid. They are a new phenomenon arising from excessive solar and wind capacity which normally have priority. (Hydro as a renewable source can be turned off without loss of potential energy, which remains available on demand as a dispatchable source.)

System costs

As the role of renewable sources increases, there has been more attention paid to system effects relating to the interaction of variable renewables with dispatchable technologies. System effects refer to the costs above plant-level costs to supply electricity at a given load and level of security of supply. A 2012 OECD Nuclear Energy Agency report focused on "grid-level system costs", the subset of system costs mediated by the electricity grid, which include a) the costs of extending and reinforcing transport and distribution grids as well as connecting new capacity, and b) the costs of increased short-term balancing and maintaining the long-term adequacy and security of electricity supply.

The report showed that while all technologies generate system costs, those of dispatchable generators are at least an order of magnitude lower than those of variable renewables. If the system costs of variable renewables were included at the level of the electricity grid, the total costs of electricity supply increased by up to one-third, depending on country, technology and penetration levels. While grid-level system costs for dispatchable technologies are lower than US$ 3 /MWh, they can reach up to $40 /MWh for onshore wind, up to $45 /MWh for offshore wind and up to $80 /MWh for solar. In addition, the greater the penetration of intermittent renewables, the higher the system costs. Introducing renewables up to 10% of total electricity supply will increase MWh costs 5% to 50% (depending on country) and typically 13-14%, but with 30% renewables the MWh costs will typically increase by one-third.

Currently, such grid-level costs are simply absorbed by electricity consumers through higher network charges, and by the producers of dispatchable electricity in the form of reduced margins and lower load factors. Failing to account for system costs means adding implicit subsidies to already sizeable explicit subsidies for variable renewables. As long as this situation continues, dispatchable technologies will increasingly not be replaced as they reach the end of their operating lifetimes, thereby seriously diminishing security of supply. Meanwhile their economic viability is seriously eroded, with

the effect most marked on those technologies with the highest variable costs. Maintaining high levels of security of electricity supply in decarbonising electricity systems with significant shares of variable renewables will require incentives to internalise the system costs, as well as market designs that adequately cover the cost of all dispatchable power production, including low-carbon nuclear energy.

The NEA report concluded that nuclear power will fare relatively better than coal or gas in the short run due to its low variable costs. In the long run, however, when new investment decisions need to be made, reduced load factors will disproportionately penalise technologies with high fixed costs such as nuclear, due to reduced capacity utilisation. In systems that currently use nuclear energy, the introduction of variable renewables is therefore likely to lead to an increase in overall carbon emissions due to the use of higher carbon-emitting technologies such as gas as back-up (notwithstanding the short-term impact on its viability).

The existence of high system costs implies that significant changes will be needed to enable an economically viable coexistence of nuclear energy and renewables in increasingly decarbonised electricity systems. Such changes may include more widespread use of carbon pricing, long-term power supply contracts, and capacity payment mechanisms in order to provide adequate incentives for new investment.

The NEA report makes four recommendations:

- Increase transparency of generation costs at system level to enable rational policy.
- Prepare regulatory frameworks which minimise system costs and internalise them for each technology so as to enable viable, adequate and sustainable supply with system balance.
- Recognise the value of dispatchable low-carbon technologies and reform energy markets to sustain them.
- Develop more flexibility in the system with load following, storage, demand management and interconnection.

Vulnerability

The vital role of transmission infrastructure gives rise to concerns about its vulnerability to hostile nation or terrorist attacks, especially from a high-altitude electromagnetic pulse (EMP). The USA could be blacked out for as long as 18 months by terror attacks on nine vital transformer substations, according to a Federal Energy Regulatory Commission (FERC) study. FERC is responsible for security regulation of interstate power grids, including blackstart recovery plans which are required for all parts of the US grid. FERC and North American Electric Reliability Corporation issued a detailed assessment of blackstart plans in 2016, based on a survey of nine unnamed grid operators, including a generating company, transmission operators and control coordinators.

Congress appointed the EMP Commission to assess the situation and recommend preventive measures. It ran tests of EMP pulse damage on grid equipment and reported in 2008 that many control systems were vulnerable. A US bill, the Critical Infrastructure Protection Act, awaits passage in the House. A new analysis of the EMP threat by the Electric Power Research Institute (EPRI), requested by the Department of Energy (DOE) and due in 2018, and an updated threat analysis by the EMP Commission will clarify options. Other nations and the EU are considering these vulnerabilities also, with South Korea reported to be in the lead regarding EMP protection. Threats range from a high-altitude EMP, which would cause deep and widespread damage, to rifle fire on a transformer substation, and cyber attacks.

Some US utilities have begun protecting their systems against EMP. Dominion Energy in Virginia plans to spend up to $500 million by 2020 to harden its system against attack, including construction of an $80 million operations centre shielded against EMP waves. Duke Energy has a project to protect three of its generation plants in the Carolinas. In the case of an EMP attack, Duke's hydroelectric plant at Lake Wylie on the state border would be available to power up from blackstart.

Particular countries

China is developing a very sophisticated grid system, since its main coal deposits are in the north, its main wind potential in the far west and its nuclear plants are on the coast – close to load centres. The grid system run by the State Grid Corporation of China (SGCC) and China Southern Power Grid Co (CSG) is growing rapidly, utilising ultra high voltage (1000 kV AC from 2009, and 800 kV DC from 2010) transmission. By 2015 SGCC invested CNY 500 billion

($75.5 billion) to extend the UHV grid to 40,000 km. By 2020, the capacity of the UHV network is expected to be some 300-400 GW, which will function as the backbone of the whole system, connecting six regional clusters. By 2020 there should be 400 GWe of clean energy sources connected, of which hydropower will account for 78 GWe, and wind power from the north a further significant portion. Wind capacity by 2020 is planned to be over 100 GWe. However, in 2015 some 34 TWh of wind output – about 20% – was forgone because of inadequate grid connections, according to the National Energy Administration.

At the end of 2009, China had budgeted to spend $600 billion upgrading its grid. Over 2014 to 2020 high-voltage transmission lines are expected to increase from 1.15 million circuit km to 1.6 million circuit km, in line with a substantial increase in generation capacity, and operational transmission losses are expected to be 5.7%, down from 6.6% in 2010. SGCC is also building export projects – see Brazil below.

A UHV DC link from Yunnan to Shenzhen in Guangdong is almost 2000 km and cost CNY 22 billion ($3 billion) for CSG, and will transmit 20 TWh/yr from 2017. It is one of 11 major transmission line projects.

The northern part of **India** suffered two massive grid failures in July 2012 leaving first 390 million people without power, and a day later, some 680 million in 22 states, highlighting the country's infrastructure challenges. The Northern grid was the first affected, then this plus parts of the Eastern and North-eastern grids, after low voltage in one place tripped a link and this led to cascade tripping. Most of the under-frequency relays (UFRs) in the northern region did not work, and load dispatch centres did not respond to the problem. Power to some essential services was restored after a few hours each time, but others were out for more than a day. All five grids are controlled by the Power Grid Corporation, which operates 95,000 km of transmission lines. The country has 33 state load dispatch centres (SLDCs), five Regional load dispatch centres (RLDCs), and a national load dispatch centre.

The USA has a patchwork of grids which are often barely interconnected. The Western Interconnection includes about 11 states plus British Columbia and Alberta. ERCOT includes most of Texas, and Eastern Interconnection takes in the rest of USA and Canada. There is very little grid capacity in the middle of the country. Exelon has temporarily curtailed off-peak output at one or more of its nuclear plants in Illinois numerous times for more than a year because of grid constraints in the PJM interconnection area.

The company has previously said intermittent grid congestion has been occurring in the region around those plants because of transmission line outages for scheduled maintenance, large influxes of wind-generated power into the grid during off-peak hours, or a combination of those factors.

In 2012 a report from the American Society of Civil Engineers said that ageing equipment and lack of capacity was leading to intermittent failures, and said that an extra $107 billion investment was needed by 2020. This may be conservative. In September 2011 a simple error led to a cascading and uncontrolled failure which affected southern California and was the most extensive in the state's history. It rivalled the 2003 failure which left much of the northeast and 50 million people without power. Among the four main causes for the Northeast blackout that investigators listed six months later: The primary utility "did not recognize or understand the deteriorating condition of its system." Over 1965 to 2009, there were 57 major grid failures in the USA and Canada, according to a study by the Institute of Electrical and Electronics Engineers, 41 of those in the USA and two of them shared.

A 2011 MIT report said that the US grid faced "a number of serious challenges over the next two decades, while new technologies also present valuable opportunities for meeting these challenges." Incorporating more renewable generation is one challenge, increased penetration of electric vehicles is another. But "the diversity of ownership and regulatory structures within the U.S. grid complicates policy-making, and a number of institutional, regulatory, and technical impediments remain that require action." It made recommendations accordingly.

The 2017 American Society of Civil Engineers Infrastructure Report Card found that most US electric transmission and distribution lines were built between 1950 and 1969, with expected operating life spans of 50 years. A May 2017 survey by the Smart Electric Power Alliance (SEPA) and Black & Veatch indicated that investment in transmission and distribution was increasing rapidly, driven partly by the need to integrate renewables. In August 2017 the Department of Energy (DOE) published a report on Electricity Markets and Reliability which recommended that the Federal Energy Regulatory Commission (FERC) take a leading role in ensuring effective grid connections to meet base-load demand more widely and reliably, as well as ensuring resilience in the grid system.

In August 2014 a new 530 km, 1000 MWe capacity HVDC line, the Champlain Hudson Power Express (CHPE) gained final approval

and will be installed underground and underwater, originating at the Canadian border in Quebec and running the length of Lake Champlain and through parts of the Hudson, Harlem, and East Rivers to New York City. Cost is put at $2.2 billion, with completion by early 2018. It is seen as hardened infrastructure, providing 1 GWe that is immune to natural disasters. In December 2016 the New England Clean Power Link was approved, a 1000 MW HVDC line over 246 km from Canada to Vermont, two-thirds of it under Lake Champlain. Hydro-Québec is offering 3000 MWe surplus capacity to US markets.

Another HVDC project, the 1000 MWe Lake Erie CleanPower Connector, will bring Ontario power 105 km to the PJM grid supplying 13 states in northeast USA at a cost of about $1 billion.

Eversource's $1.6 billion Northern Pass Transmission Project is proposed to bring 1090 MWe of Quebec hydropower to New Hampshire and Massachusetts. It is 380 km, 320-kV DC, but in 2018 failed to get approval from New Hampshire where about one-third of its length would be underground.

In November 2013 the 1500 MWe Gateway project was approved by the US government, a 1600 km HVAC link from southern Wyoming to southern Idaho to strengthen the western grid and transmit wind power west to a more populous area.

In 2015 the $2 billion Clean Line Energy proposal was being put forward for a 1129 km 3500 MWe HVDC transmission link from wind farms in Oklahoma and Texas to Memphis, Tennessee, linking with the TVA grid. Construction of this Plains & Eastern Clean Line was due to start in 2017 for 2020 commissioning, with GE supplying the HVDC converter stations. Arkansas initially opposed the project, and then Missouri rejected it in August 2017 pending consent from affected counties. This would be the first of several projects to link wind generation in the middle of the continent to markets east and west. The proposed Grain Belt Express Clean Line will be 1250 km HVDC from western Kansas across northern Missouri and Illinois, connecting with the PJM Interconnection market. Other proposals are for links from wind farms in northern Texas and western Kansas.

In 2014 ERCOT's Competitive Renewable Energy Zones (CREZs) came into operation, linking 11.6 GWe of wind generation in north Texas panhandle and west Texas with load centres in the south, with 5700 km of 345 kV transmission lines. It is designed to accommodate 18.5 GWe. The wind generation is backed up by a large fleet of natural gas combined cycle units.

In mid-2016 the California Independent System Operator (CAISO) said in a 700-page report that expanding its operations to include more western states would result in a more efficient electricity grid, reduce greenhouse gas emissions across the west, as well as meet or exceed the state's goal of obtaining half of its energy from renewable sources. The study showed that an 11-state regional market would cut costs by allowing generators to sell excess electricity more easily across state lines, as well as letting California import larger amounts of renewable energy from neighbouring states. CAISO said that California was set to produce a surplus of 13 GWe of renewable energy by 2025 that would have to be shut down when peak generation exceeded demand. Expanding the ISO territory would enable this to be shared or dumped interstate without shutting down turbines.

In Germany the existing north-to-south to lines are overloaded and incapable of transmitting sufficient wind-generated power from the north to replace closed capacity in the south. In May 2011 the German federal network agency and grid authority, Bundesnetzagentur (BNetzA), reported on the implications of plans to close down nuclear generation and greatly ramp up the contribution of wind and solar sources. It strongly warned of resulting vulnerability to major failures and also unreliability especially in the south. Grid stability was the major concern, along with generation and transmission capacity.

In December 2012 a report from the German Energy Agency (Deutsche Energie-Agentur GmbH, DENA) showed that investment of €27.5 billion to €42.5 billion was required by 2030 to expand and upgrade electricity distribution to cope with increased renewables share of supply. The DENA Distribution Study said that grid and distribution expansion of 135,000 km to 193,000 km was needed. It called for reforms to the regulatory framework to help network operators realise the returns needed as incentives for the necessary investments. DENA is 50%-owned by federal government ministries and 50% by German financial institutions. The Distribution Study was supported by German generation and grid companies including EnBW, EOn and Vattenfall.

In October 2015 the government approved plans for four major high-voltage DC transmission lines totalling about 1000 km from the north, and close to populated areas, to be built underground, initially due to Bavarian opposition to overhead lines. The energy ministry estimated that the underground option would cost €3 to €8 billion more than overhead lines, to be added to consumers' bills, but was expected to speed up approvals. In May 2016 BNetzA put the cost of the required 7000 km of new transmission lines at €35

billion, with priority given to the three north-south links by 2022 when the last nuclear plant is due to close. Plans for these four north-south HVDC corridors are behind schedule.

As well as 2800 km of new lines – jointly called SuedLink – about 1500 km of the present grid is being upgraded. State governments have agreed to let BNetzA coordinate plans, rather than asserting regional interests. One project under construction is the 380 kV so-called Thuringer Bridge transmission link connecting Saxony-Anhalt in eastern Germany with Bavaria and due to be completed early in 2016. A further increase in north-south transmission capacity is planned by converting the 400 km North Rhine-Westphalia to Baden-Württemberg line to 2 GW HVDC. This was due to be commissioned in 2019 when EnBW's 1392 MWe Phillipsburg 2 nuclear plant is due to close, but is about a year behind.

Plans for a 1400 MWe HVDC link with Norway promise to help Germany's renewable energy objectives, as the interconnector between Denmark and Norway has been doing for Denmark's wind energy for some years. (Denmark's wind turbines depend heavily for their effective utilization on 29 GWe of hydro capacity in Norway, over 1 GWe of which can be dispatched promptly when wind power is unavailable in West Denmark. This will rise by 700 MW in 2014. Hence, there is a natural interdependence between West Denmark's wind and Norway's hydro. With good winds, power can be exported back to Norway and there conserve hydro potential.)

The Czech Republic is one of the adjacent countries affected by Germany's grid problems. Since mid-2012 the 2 GWe Temelin plant has operated about 100 MWe below capacity as instructed by grid operator CEPS because of grid security issues caused by power surges due to renewable power production in Germany. The Czech Republic and Poland have installed phase-shifting transformers on their German border to block German electricity dumping; France Netherlands and Belgium already had them.

Austria is another country experiencing difficulties due to input from subsidized wind and solar PV. The Austrian Power Grid (APG) is having increasing difficulty in balancing unpredictable supply to demand requirements. This has raised the need for adequate sourcing of balancing power, requiring dependable sources such as gas-fired generating units to be available. In Austria most of these are now out of service, unable to compete economically, and hence the country has high reliance on uncertain German supply. APG proposes capacity payments to keep fossil fuel capacity in standby

mode, especially as further wind capacity comes on line with limited grid access.

France's grid operator RTE plans to invest €15 billion ($19 billion) on the nation's grid by 2020, and a further €20 billion by 2030 with the present energy mix. However, it says that €50 billion would be required by 2030 to cope with a nuclear share reduced from 75% to 50% of supply and replace this with renewables. The basic grid investments are needed to improve security of supply and accommodate rising renewable power capacity. RTE has 105,000 km of transmission lines and grid transport costs account for about 10% of consumers' bills.

France already exports a lot of electricity to Italy. In 2015 RTE started work on the new 1200 MWe HVDC Savoie-Piemont connection from Chambery to Turin in Italy, costing about €1 billion, which may have relevance to the new energy policy and domestic supply cap. It will be the longest subterranean high-voltage power line (320 kV) when it goes into service in 2019. In 2014 Italy imported 19 TWh through existing 2700 MWe links, and the new connection will add capacity for 10.5 TWh more.

Italy's Terna is the TSO, with 64,000 km of transmission lines. It will share the cost of the HVDC Savoie-Piemont connection.

Ukraine's new government formed in 2014 aims to integrate with the European power grid and gas network to make the country part of the European energy market by 2017. In March 2015 an agreement was signed by Ukraine's Ukrenergo distribution company and Polenergia, a Polish counterpart, to export electricity as part of the Ukraine-EU 'energy bridge', and related to the Baltic Energy Market Interconnection Plan. This will enable greater use of Ukraine's nuclear capacity and is to generate funds to pay for increasing that capacity at Khmelnitski. A 750 kV transmission connection from Khmelnistki to Rzeszow in Poland is envisaged, taking in also Ukraine's Burshtyn Island coal-fired plant, with Khmelnistki unit 2 being disconnected from the Ukraine grid. In June 2015 the government approved the project.

Russia's Federal Grid Company is 80% owned by the government and controls 125,000 km of transmission lines over 13.6 million sq km. Its customers are regional distribution companies ('discos'), electricity suppliers and large industrial enterprises.

Japan's grid is very unusual in that on the main island, Honshu, the northeastern half including Tokyo is 50 Hz, served by Tepco (and Tohoku), the southwestern half including Nagoya, Kyoto and Osaka is 60 Hz, served by Chubu (with Kansai & Hokuriku), and there is

only 1 GWe of frequency converters connecting them. This arises from original equipment coming from Germany and USA respectively. The interconnection is being increased to 2.1 GWe, funded by the utilities. Early in 2013 it was announced that METI will establish a new body to balance electricity supply and demand in wide areas across Japan, as early as 2015. The new body will manage grid and transmission facilities, which are currently owned and managed by utility companies.

Between **Finland and Sweden**, the Fenno-Skan 2 HVDC link was completed in December 2011, increasing the connection by 40%. This improves the functioning of the Nordic market and enables Finland to import its electricity shortfall from Sweden rather than Russia. It is 300 km, two-thirds submarine across the Gulf of Bothnia, and 800 MWe at 500 kV DC. It cost €315 million. Fingrid is planning a further connection with Sweden, to be in place by 2024.

In **Brazil**, a 2084 km link is being built by the State Grid Corporation of China (SGCC), from the 11,233 MWe Belo Monte hydro plant on the Xingu River in northern Para state to the southern economic centres in Minas Gerais state. It is the first such UHV export project for the company, and is 800 kV DC. In addition State Grid Brazil is building a 250 km UHV link from Bel Monte power plant to near Rio de Janeiro. The two projects are expected to cost $4.7 billion. SGCC is already the fourth largest TSO in Brazil.

Major regional grid projects

Baltic Energy Market Interconnector Plan (Baltic grid map pdf)

The planned Visaginas nuclear plant is envisaged as the cornerstone of the Baltic Energy Market Interconnector Plan (BEMIP) linking to Poland, Finland and Sweden. A high-voltage (400 kV) 1000 MW DC southwest interconnection – PowerBridge or LitPol Link – costing €250-300 million, to improve transmission capacity between Lithuania and Poland is to be built, with 500 MW by 2015 and another 500 MW planned by 2020. Much of the funding is from the European Union (EU), and work is ahead of schedule. Further transmission links between Estonia and Latvia will be needed for the three Baltic states to synchronise with Poland and EU by 2025.

This follows inauguration of an interconnector between Estonia and Finland to the north – Estlink-1, a 150 kV, 350 MW HVDC cable costing €110 million and also supported by EU funding. The 170 km 450 kV HVDC Estlink-2 further east and now under construction

will provide a further 650 MW early in 2014. The project is budgeted to cost around €320 million, which will be divided between TSOs Finngrid and Elering (Estonia), with €100 million to be provided by the EU as part of the EU's extensive economy recovery package. Both will be operated by the two TSOs.

Another major transmission link westward under the Baltic Sea, the 700 MWe NordBalt 300 or 400 kV HVDC project, is planned between Klaipeda in Lithuania and Nybro in Sweden (400 km), by Svenska Kraftnat and LitGrid. The €550 million project is expected to be completed by 2016. (The Baltic states and Belarus have good interconnection of grids from the Soviet era, but this did not extend to Poland, let alone to Germany. Kaliningrad gets all of its electricity from Russia, via the Lithuanian grid.)

Lithuania's revised energy policy in 2012 involves rebuilding the grid to be independent of the Russian system and to work in with the European Network of Transmission System Operators (ENTSO) synchronous system, as well as strengthening interconnection among the three Baltic states.

This EU integration was an important factor leading to Russia suspending work on its new Baltic nuclear power plant in its exclave of Kaliningrad. It was designed for the EU grid and about 20% built. Despite endeavours to bring in west European equity and secure sales of power to the EU through proposed transmission links, the 1200 MWe plant is isolated, with no immediate prospect of it fulfilling its intended purpose. Kaliningrad has a limited transmission link to Lithuania, and none to Poland, its other neighbour. Both those countries declined to buy output from the new Baltic plant. Lithuania does not wish to upgrade its Kaliningrad grid connection to allow Baltic NPP power to be sent through its territory and Belarus to Russia. As well as upgrading the Lithuania link, Russian grid operator InterRAO had plans to build a 600-1000 MWe link across the Kaliningrad border to Poland and a 1000 MWe HVDC undersea link to Germany, but with no customers these plans are not proceeding. In March 2013 Rosatom said that Russia had applied for Kaliningrad to join the EU grid system (ENTSO-E), evidently without response.

European and Scandinavian power exchanges

Several power exchanges exist in the European area: NordPool, covering Scandinavia, the Baltic states and Poland; European (EEX), covering France, Germany Austria and UK; GME covering Italy, Switzerland and some countries east of Italy; and OMEL for Spain and Portugal. These trade on the spot and futures markets.

North Sea grid

In a move towards meeting the EU's goal of reaching a 20% share of energy from renewable sources by 2020, nine European countries have agreed to build a power grid of high-voltage cables under the North Sea. It would be the first multinational grid designed to address the fluctuating nature of 'green' power generation. The North Sea Grid Initiative includes Germany, Denmark, Norway, Sweden, Belgium, France, Netherlands, Luxembourg, and the United Kingdom.

The project aims to link up some 100 GW of offshore wind power which is currently planned by European power companies. The UK has launched a £100 billion programme to boost its offshore wind farms; already the world's biggest at around 1 GW, to as much as 40 GW by 2020. The project will have an estimated cost of about $40 billion, and is expected to be operational by 2023, balancing supply and loads among regions and from large wind and solar farms.

In February 2016 there were a number of subsea cable projects under construction or just completed in Europe:

Skagerrak 4, 700 MW connecting Norway and Denmark, commissioned March 2015.
NordBalt, 700 MW connecting Sweden and Lithuania, due 2016.
Western HVDC link, 2200 MW connecting Scotland and Wales, due 2017.
MON.ITA, 1000 MW connecting Italy and Montenegro, due 2019.
NEMO, 1000 MW connecting UK and Belgium, due 2018.
Nord.link, 1400 MW connecting Germany and Norway, due end 2020.
UK-Norway NSN, 1400 MW connecting UK and Norway, due 2021.
IFA 2, 1000 MW connecting UK and France (proposed), by 2020.
FABlink, 1000-1400 MW, connecting UK and France (proposed), by 2022.

Construction of the 1.4 GW North Sea Link between Norway and Northumberland in the UK passed the halfway point and was heading for completion by 2021, UK transmission company National Grid said in June 2020. Another 1.4 GW link to Scotland, Northconnect, is planned once North Sea Link and Nordlink (to Germany) are operational.

Mediterranean links

A 1.4 GWe link between Spain and Morocco has been in operation since 1998.

The new 600 MWe DC Elmed link is planned to connect the Italian grid at Partanna on Sicily to El Hawaria in Tunisia from 2025. The subsea cable length is about 192 km, with 32 km of underground cable in Sicily and 5 km in Tunisia. Cost estimate is €600 million, half financed by the EU.

East Asian grid

Korea Electric Power Corporation (Kepco) is promoting a plan to connect Busan in South Korea with Fukuoka in the south of Japan, via Tsushima Island. This would involve a 50 km section to the island and another 150 km to Japan, and would enable South Korea's anticipated power surplus to alleviate Japan's shortage. This would be a 60 Hz connection with that part of Japan.

This follows the 2012 proposal by Japan's Softbank for an Asia Super Grid connecting Korea, China, Japan, Russia (Vladivostok and Khabarovsk) and Mongolia. Softbank is reported to have joined with Newcom in Mongolia to develop a 300 MWe wind farm in the Gobi desert to supply Japan eventually. Further plans are for up to 7 GWe there. Newcom already supplies 5% of Mongolia's power from wind.

Southern African Power Pool (SAPP)

SAPP coordinates the power systems of 12 Southern Africa Development Community (SADC) countries (Angola, Botswana, the Democratic Republic of Congo, Lesotho, Malawi, Mozambique, Namibia, South Africa, Swaziland, Tanzania, Zambia, and Zimbabwe). Nine of the countries are so-called 'operating members', meaning they are linked to the interconnected grid that carries around 97% of energy produced in the SAPP. Total installed capacity in 2014 was 57 GWe of which less than 52 GWe was available. Most of the electricity is generated in South Africa, with 77% of the capacity. Demand exceeds supply. The World Bank has offered $20 million to fund regional energy projects under SAPP.

In August 2015 SADC announced that 24 GWe of capacity was under construction to come on line by 2019, about 70% of it from renewable sources and the rest from the big Medupe and Kusile coal-fired power stations in South Africa. The largest project was the first stage of the Grand Inga hydro power plant on the Congo River in DR Congo, which might eventually produce 44 GWe.

Eastern African Power Pool (EAPP)

The World Bank is funding a new Eastern Electricity Highway project to connect Ethiopia with Kenya and eventually with the

Southern African Power Pool. It is the first phase of a $1.3 billion Eastern Africa power integration program, with $243 million for Ethiopia and $441 million for Kenya from the Bank, which says that the "project will change the fundamentals of the power sector in East Africa". A 400 kV AC, 2000 MWe link between Kenya and Tanzania was funded by the African Development Bank early in 2015.

Ethiopia is planning to increase hydropower capacity to raise capacity from 2.4 to 10 GWe and become a regional electricity exporter. State-owned power utility Ethiopian Electric Power has signed a US$120 million contract with China Electric Power Equipment and Technology to construct a 433 km high-voltage transmission line from Wolaita in the south of the country to the Kenyan border. This 500 kV, 2000 MW HVDC link to Kenya is due to be completed in 2018, funded by the World Bank.

West Africa Power Pool (WAPP)

The Economic Community of West African States (ECOWAS) earlier resolved to implement a West Africa Power Pool (WAPP). In July 2015 an agreement was signed by several countries for cooperation in the development of an integrated West African regional nuclear power programme, related to this.

Central and South America

The world's longest HVDC link, 2400 km, was commissioned in Brazil in 2014 to bring 3150 MW of power from two hydropower plants in the northwest to Sao Paulo. Brazil, Argentina, Uruguay and Paraguay with shared large hydro projects already have extensive grid connection.

Chile, Colombia, Ecuador and Peru are seeking to integrate their power systems through the Andean Electrical Interconnection System (SINEA) project. In 2015 Bolivia, with Argentina, Brazil, and Paraguay, agreed to invest more than US$ 620 million into an electricity interconnection program to result in 1400 km of grid infrastructure. Then Bolivia agreed with Peru for interconnection.

In Central America, driven by renewable energy projects, the final link of the Central American Electrical Interconnection System (SIEPAC) was completed in 2014, connecting six countries from Guatemala to Panama via an 1800 km line.

Australia

Eastern Australia's National Electricity Market (NEM) operates the world's largest interconnected power system that runs for more than 5,000 kilometres from North Queensland to Tasmania and central South Australia, and supplies some $10 billion electricity annually to meet the demand of more than 10 million end users.

Smart grids

'Smart grid' refers to a class of technology for electricity delivery which uses computer-based controls to monitor and match supply with real-time end-user demand, varying prices accordingly. It involves two-way communication between distributor and customers' meters and switchboards, with the management of that information to optimise efficiency. A key feature of the full smart grid is automation technology that allows the utility to adjust and control each individual device or millions of devices from a central location. Smart grids allow optimal integration of domestic-scale renewables into the grid, and also integration of electric vehicles into the system.

Smart grids have major implications at the distribution level, but little at the TSO level. Some 80% of smart grid investment is at DSO level, very little at TSO. Despite talk of electricity highways, HVDC etc, most non-hydro renewable sources are connected to low-voltage distribution networks rather than the high-voltage grids.

Impediments to improvement

The high cost of transmission projects is one disincentive to investment in new capacity.

Acquiring and managing the right of way for transmission assets is a complicated and cumbersome process in many countries, with reliability and customer opinion at stake. Electric utilities and TSOs must manage numerous and often competing interests as they negotiate easements for transmission projects. These will be driven by reliability and capacity objectives, but landowners and government officials have other priorities and interests.

In France, opponents of the 163-km Cotentin-Maine project connecting the new Flamanville reactor to the main grid argued

that uncertainty over the safety of living near high-voltage power lines, including the risk of causing leukemia in children, meant that the project should not proceed. Opponents included environmental groups and local public associations. The country's top administrative court rejected the appeal, saying it was a public interest project and that enough safety assessments had been carried out.

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